



HW Testing

Electronics Projects

Abstract

This document serves the purpose of planning, testing and documenting the developed hardware in the project module 3.

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1. Test procedure

After the hardware has been completely soldered and assembled, it is of utmost importance to carefully inspect the solder joints under a microscope. This pre-testing step can help prevent potential damage to the electronics caused by shorts or misplaced components.

Next, the “life signs” test can begin, where the board is powered on for the first time. During this step, all safety considerations mentioned in the safety chapter must be observed. All internal voltages and current draws should be documented and compared against the expected values. In case of a failure, a structured debugging approach must be followed. If the life signs test passes, it is permissible to proceed to the basic hardware functionality tests.

The basic hardware functionality test aims to verify the operation of sub-circuits groups. It follows a systematic approach, starting from the audio input connectors, through the analog input circuitry, codec, analog output circuitry, and ending at the output connectors.

2. Safety

Although not part of the testing, care must be taken when soldering and handling the soldering iron. Soldering temperatures range from 300 °C to 400 °C and can cause serious burns if not handled properly.

When powering the board for the first time, it is good practice to use a laboratory power supply with a current-limit feature to prevent severe damage in case of a short circuit. Even better results can be achieved by powering the board at a slightly lower voltage initially to further limit power in case of a fault.

If abnormal behavior such as smoke, overheating, or excessive current draw is observed, disconnect power immediately and inspect the hardware before continuing.

Work on a ESD protected surface to prevent ESD damage to components.

3. Test Plan

Test item, Description	Test conditions	Expected, Tolerance	Measured	Comparison, verdict	Remarks, Discussion
General setup for all the following tests (unless stated otherwise): - Power in by laboratory power supply, 5V - +3V0 (P2, Pin 1) connected to +3V3					
Voltage at +3V3 ESP supply	No input signal	$3.32V \mid 33.2k\Omega (\pm 1\%) \cdot 100\mu A (\pm 1\%) = 3.25V \dots 3.39V$	3.29V	In range → passed	
Voltage at +3V3 digital supply (VD)	No input signal	$3.32V \mid 33.2k\Omega (\pm 1\%) \cdot 100\mu A (\pm 1\%) = 3.25V \dots 3.39V$	3.29V	In range → passed	
Voltage at +5V analog supply (VA)	No input signal	$4.99V \mid 49.9k\Omega (\pm 1\%) \cdot 100\mu A (\pm 1\%) = 4.89V \dots 5.09V$	4.99V	In range → passed	
Voltage at DMX output (DMX_+5V)	No input signal	$5.4V \mid 270\Omega (\pm 1\%) \cdot 20mA (\pm 1\%) = 5.29V \dots 5.51V$	5.24V	Out of range, considered acceptable ^{*)} → passed	*) Taking into account the tolerance envelope
Voltage at Boost output	No input signal	$5.51V \mid \left(\frac{301k\Omega (\pm 1\%)}{30k\Omega (\pm 1\%)} + 1 \right) \cdot 500mV (\pm 15mV) = 5.25V \dots 5.78V$	5.67V	In range → passed	
Ripple on +3V3 ESP supply	No input signal	$0.8\mu V_{RMS} + 1mV \cdot 10^{-\frac{50dB}{20}} \rightarrow < 0.80\mu V$	4.0mV _{pp}	Out of range, considered acceptable ^{*)} → passed	
Ripple on +3V3 digital supply (VD)	No input signal	$0.8\mu V_{RMS} + 1mV \cdot 10^{-\frac{50dB}{20}} \rightarrow < 0.80\mu V$	4.4mV _{pp}	Out of range, considered acceptable ^{*)} → passed	*) Oscilloscope has not enough accuracy, measures above the expected value even when no signal is connected
Ripple on +5V analog supply (VA)	No input signal	$0.8\mu V_{RMS} + 10mV \cdot 10^{-\frac{50dB}{20}} \rightarrow < 0.81\mu V$	1.5mV _{pp}	Out of range, considered acceptable ^{*)} → passed	
Ripple on DMX output (DMX_+5V)	No input signal	$< 40mV_{pp}$	6mV _{pp}	In range → passed	
Ripple on Boost output	No input signal	$\frac{1A \cdot 0.18}{1MHz \cdot 10\mu F} \rightarrow < 18.3mV$	5mV _{pp}	In range → passed	
Current on Power in at 5 V	No input signal	$< 180mA$	173mA	In range → passed	
Current on Power in at 5V, +3V0 disconnected	No input signal, +3V0 disconnected	$< 160mA$	128mA	In range → passed	
Current ripple on Power in at 5 V	No input signal	$\frac{1mV}{5V/180mA} \rightarrow < 36\mu V$	1mA _{pp}	Out of range, considered acceptable ^{*)} → passed	*) Measurement method provides not enough accuracy
Microphone voltage	No input signal	$2.6V \mid 2V \dots < 5V$	2.48V	In range → passed	
Signal Peak finger snip	No input signal, finger snip next to the microphone	$50mV \mid 0V \dots 2.5V$	68mV _{pp}	In range → passed	
Signal Peak finger on microphone	No input signal, finger on microphone	$> 300mV \mid 0V \dots 2.5V$	500mV _{pp}	In range → passed	Note: Expected value given as amplitude, measured peak-to-peak, conversion needed before comparison
Signal Peak finger snip after Preamp	No input signal, finger snip next to the microphone	$> 2V \mid 0V \dots 2.5V$	5V _{pp}	In range → passed	
Signal Peak finger on microphone after Preamp	No input signal, finger on microphone	$> 2V \mid 0V \dots 2.5V$	5V _{pp}	In range → passed	

Preamp reference voltage (U4)	No input signal	$2.5V \mid 5V \cdot \frac{100k\Omega \pm 1\%}{(100k\Omega \pm 1\%) + (100k\Omega \pm 1\%)} = 2.475V \dots 2.525V$	2.49V	In range → passed	
Line in signal after Anti-Aliasing filter, left channel	Input signal: 500mVpp 1kHz sine	No attenuation @ 1kHz → 500mV_{PP}	532mV_{PP}	Acceptable with tolerance → passed	
Line in signal after Anti-Aliasing filter, right channel	Input signal: 500mVpp 1kHz sine	No attenuation @ 1kHz → 500mV_{PP}	532mV_{PP}	Acceptable with tolerance → passed	
Voltage at DMX A output	No input signal	$3V + 5V/2 = 5.5V$	4.46V	Not as expected, considered acceptable ^{*)} → passed	*) Calculated with max values from the datasheet, is in range for $V_{OD,min} = 1.5V$
Voltage at DMX B output	No input signal	$3V - 5V/2 = 0.5V$	0.78V	Not as expected, considered acceptable ^{*)} → passed	
General setup for all the following tests (unless stated otherwise): - Power in by laboratory power supply, 5V - +3V0 (P2, Pin 1) connected to +3V3 - Wire loop SDOUT (P1, Pin 11) to SDIN (P1, Pin 16) - Reset CS4271 Codec after power applied					
Crystal Frequency	No input signal, 1:10 Probe	16.384MHz ... 25.600MHz	24.88MHz	In range → passed	
MCLK Frequency	No input signal	1.024MHz ... 25.600MHz	12.3MHz	In range → passed	
LRCLK Frequency	No input signal	4kHz ... 50kHz	48kHz	In range → passed	
SCLK Frequency	No input signal	$LRCLK \times 64 = 256kHz \dots 3.2MHz$	3.07MHz	In range → passed	
SDOUT signal present	No input signal	Signal present	Yes	As expected → passed	
Output signal after Reconstruction filter, left channel	Input signal: 1mVpp 1kHz sine	No attenuation @ 1kHz → 500mV_{PP}	533mV_{PP}	Acceptable with tolerance → passed	
Output signal after Headphone Amplifier, left channel	Input signal: 1mVpp 1kHz sine	Amplification factor = 5 → 2.5V_{PP}	2.62V_{PP}	Acceptable with tolerance → passed	
Output signal after Reconstruction filter, right channel	Input signal: 1mVpp 1kHz sine	No attenuation @ 1kHz → 500mV_{PP}	536mV_{PP}	Acceptable with tolerance → passed	
Output signal after Headphone Amplifier, right channel	Input signal: 1mVpp 1kHz sine	Amplification factor = 5 → 2.5V_{PP}	2.60V_{PP}	Acceptable with tolerance → passed	
Frequency sweep	Input signal: Frequency sweep 400Hz ... 20kHz	No attenuation in FFT spectrum	Yes	As expected → passed	

Information on the conducted tests:

Date: 31.10.2025, 07.11.2025 and 14.11.2025

Location: ZHAW, Technikumstrasse 71, 8400 Winterthur, Room TS04.11

Test personnel: Thomas Perri and Bogdans Grebnevs

Equipment: Keysight U3402A digital multimeter, Tektronix TDS 2012C oscilloscope, TTI EL302RT laboratory power supply

For further detailed information see the accompanying Excel table.

4. Test Analysis

During testing, several issues related to missing or incorrect electrical connections were identified and subsequently corrected.

Problem 1: Incorrect power-supply flag naming

During schematic design, some power-supply flags were inconsistently named across different sheets. As a result, the corresponding nets were not connected in the PCB layout.

Solution: Two patch wires were added: one from the 3.3 V line to C29 and another from JP5 to R27.

After applying these modifications, the power-supply functioned correctly throughout the entire device.

Problem 2: Reference-voltage (VCOM) buffer of the input filter

Similar to Problem 1, the signal flags for the VCOM reference voltage were mixed up. This caused an unintended bridging of the buffer stage and led to a short between the codec and filter VCOM pins.

Solution: The affected PCB traces were cut in three locations, and two patch wires were added: one from the codec's VCOM output (via) to the input pin of the buffer op-amp, and another from R37 to C37.

After implementing these changes, both the codec VCOM output and the buffer output showed correct levels, restoring proper operation of the input filter.

After implementing these corrections, all tests were completed successfully, with only minor deviations that stayed within acceptable limits. In summary, the DUT has passed the hardware validation and is ready to proceed to the next phase of firmware development.